Shape optimization using φ-FEM

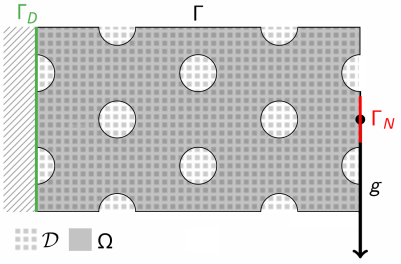
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Toy problem

Goal: minimize the compliance of an elastic cantilever beam under a **volume constraint**.



Let $\varphi : \mathcal{D} \to \mathbb{R}$ be a level-set function,
$\Omega = \{\varphi < 0\}$ be the domain to optimize,
$g \in L^2(\Gamma_N)$ be a traction force,
$u_\Omega : \Omega \to \mathbb{R}^2$ be the displacement field,
$\varepsilon(u_\Omega) = 0.5(\nabla u_\Omega + \nabla u_\Omega^T)$ be the linear elastic strain,
$\sigma(u_\Omega) = 2\mu\varepsilon(u_\Omega) + \lambda \mathrm{div}(u_\Omega) I$ be the linear elastic stress where (λ, μ) are the Lamé coefficients.

State equations:
We seek u_Ω satisfying:

$$-\mathrm{div}(\sigma(u_\Omega)) = 0 \quad \text{in } \Omega \subset \mathcal{D}, \tag{1}$$
$$u_\Omega = 0 \quad \text{on } \Gamma_D, \tag{2}$$
$$\sigma(u_\Omega)n = g \quad \text{on } \Gamma_N, \tag{3}$$
$$\sigma(u_\Omega)n = 0 \quad \text{on } \Gamma, \tag{4}$$

Shape optimization problem:

$$\min_{\Omega \in \mathcal{O}} \mathcal{J}_\Omega(u_\Omega) = J(u_\Omega) + \Lambda|\Omega|, \tag{5}$$

where $\mathcal{O}$ the set of admissible shapes,
$J(u_\Omega) = \int_\Omega \sigma(u_\Omega) : \varepsilon(u_\Omega)$,
and $\Lambda \in \mathbb{R}_{>0}$ is a material cost.
The distributed formulation of the shape derivative of $\mathcal{J}$ is [4]:

$$\mathrm{d}\mathcal{J}_\Omega(u_\Omega; \xi) = \int_\Omega S_1 : \nabla\xi + \Lambda \int_\Omega \mathrm{div}(\xi),$$

where $S_1 = 2\nabla u_\Omega^T \sigma(u_\Omega) - \sigma(u_\Omega) : \varepsilon(u_\Omega) I$.

Gradient descent algorithm

Main loop:

1. **Solve state eqs. (1) to (4):** using φ-FEM, **see I.**
2. **Compute the descent direction:** using FEM, **see II.**
3. **If:** $|\mathrm{d}\mathcal{J}_\Omega(u_\Omega; \theta)| < \mathrm{tol}$, **return:** Ω.
4. **Update the level-set:** seek ϕ s.t.
$$\partial_t \phi + \theta \cdot \nabla\phi = 0 \quad \text{in } (0, T_u) \times \mathcal{D},$$
$$\phi(0, \cdot) = \varphi \quad \text{in } \mathcal{D},$$
using FEM.
5. **Reinitialize the level-set:** seek ψ s.t.
$$\partial_t \psi + \mathrm{sgn}(\phi)(|\nabla\psi| - 1) = 0 \quad \text{in } (0, T_r) \times \mathcal{D},$$
$$\psi(0, \cdot) = \phi \quad \text{in } \mathcal{D},$$
using FEM.

Context

Optimal design seeks to **maximize the efficiency of resource use under practical constraints**. The **level-set approach** has become a widely used tool to describe complex geometries evolving during the optimization process [1].

In this work, **we couple the level-set shape optimization framework with φ-FEM**, an unfitted finite element method (FEM) based on a description of the geometry by a level-set [5].
The φ-FEM allows us to keep a **satisfactory precision** using a **fixed (cartesian) mesh** throughout the optimization process.

Our long-term goal is to derive an efficient shape optimization algorithm applied to the design of patient-specific angioplasty devices.

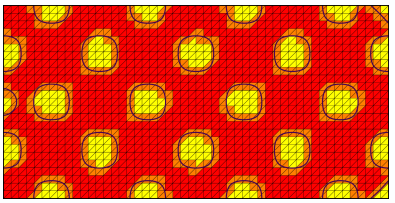
The implementation is done in FEniCSx [2] with the FormOpt [4] and phiFEM [3] python libraries.

I. Solve the state equations (1) to (4)

Let $V_h(\Omega_h)$ and $V_h(\Omega_h^\Gamma)$ be Lagrange FE spaces of vector fields over Ω_h and Ω_h^Γ resp. and $Q_h(\Omega_h^\Gamma)$ be a Lagrange FE space of tensor fields over Ω_h^Γ. Let $M_h = V_h(\Omega_h) \times Q_h(\Omega_h^\Gamma) \times V_h(\Omega_h^\Gamma)$.
φ-FEM scheme [5]: seek $(u_h, y_h, p_h) \in M_h$ s.t. $\forall (v_h, z_h, q_h) \in M_h$ we have:

$$\int_{\Omega_h} \sigma(u_h) : \varepsilon(v_h) + \int_{\partial\Omega_h} (y_h n) \cdot v_h + \gamma_u \int_{\Omega_h^\Gamma} (y_h + \sigma(u_h)) : (z_h + \sigma(v_h))$$
$$+ \frac{\gamma_p}{h^2} \int_{\Omega_h^\Gamma} \left(y_h \cdot \nabla\varphi_h + \frac{1}{h} p_h \varphi_h \right) \cdot \left(z_h \cdot \nabla\varphi_h + \frac{1}{h} q_h \varphi_h \right) + \sigma G_h((u_h, y_h), (v_h, z_h)) = \int_{\Gamma_N} g \cdot v_h.$$

where G_h is a stabilization term and $\gamma_u, \gamma_p, \sigma \in \mathbb{R}$ are parameters.



$\Omega_h = \{\blacktriangle\} \cup \{\blacktriangle\}$, $\Omega_h^\Gamma = \{\blacktriangle\}$.

Displacement field animation:

II. Compute the descent direction

Let $\Theta_h(\mathcal{D})$ be a Lagrange FE space of vector fields vanishing over $\Gamma_D \cup \Gamma_N$ and $c_1, c_2 \in \mathbb{R}_{>0}$ be given.
FEM scheme: find $\theta_h \in \Theta_h$ s.t. $\forall \xi_h \in \Theta_h$ we have:

$$\int_{\mathcal{D}} (c_1 \nabla\theta_h : \nabla\xi_h + c_2 \theta_h \cdot \xi_h) = -\mathrm{d}\mathcal{J}_h(u_h; \xi_h),$$

where $\mathrm{d}\mathcal{J}_h(u_h, \xi_h)$ is a discretization of $\mathrm{d}\mathcal{J}_\Omega(u_\Omega; \xi)$ and u_h is computed in **I.**

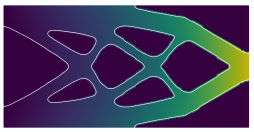
Descent direction animation:

Results and future work

Future work:

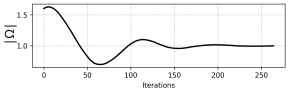
- Use the Hadamard formulation of $\mathrm{d}\mathcal{J}_\Omega(u, \xi)$ and an interface φ-FEM scheme for II to improve θ_h.
- Numerical analysis of **II** using φ-FEM.
- Validation on other problems (e.g. L-shaped, ...).
- Parallel and 3D implementations.

Long term goal:
design angioplasty balloons of optimal (variable) thickness to better match patients physiology.



u magnitude on optimized Ω.





References

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